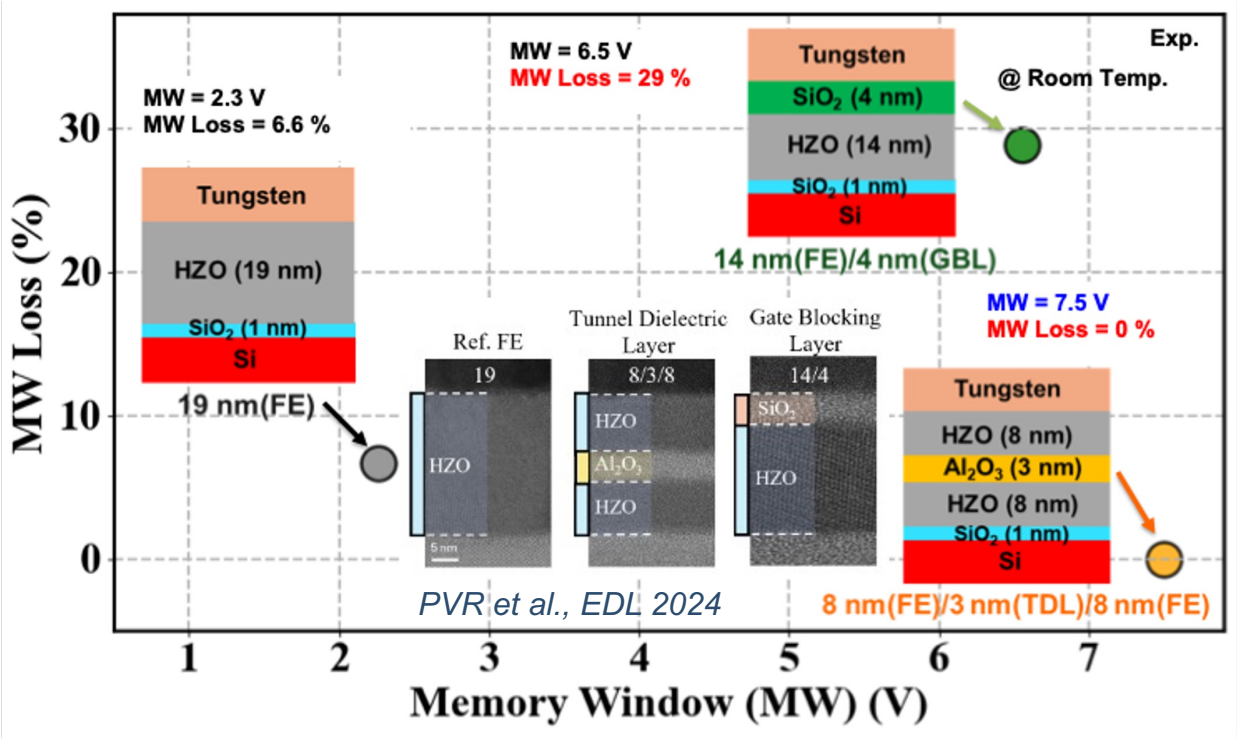

MW = 2.3 V
MW Loss = 6.6 %
Tungsten
HZO (19 nm)
SiO2 (1 nm)
Si
19 nm(FE)
MW = 6.5 V
MW Loss = 29 %
Tungsten
SiO2 (4 nm)
HZO (14 nm)
SiO2 (1 nm)
Si
14 nm(FE)/4 nm(GBL)
Exp.
@ Room Temp.
Ref. FE
19
HZO
5 nm
Tunnel Dielectric Layer
8/3/8
HZO
Al2O3
HZO
Gate Blocking Layer
14/4
SiO2
HZO
MW = 7.5 V
MW Loss = 0 %
Tungsten
HZO (8 nm)
Al2O3 (3 nm)
HZO (8 nm)
SiO2 (1 nm)
Si
8 nm(FE)/3 nm(TDL)/8 nm(FE)
PVR et al., EDL 2024
MW Loss (%)
0
10
20
30
Memory Window (MW) (V)
1
2
3
4
5
6
7